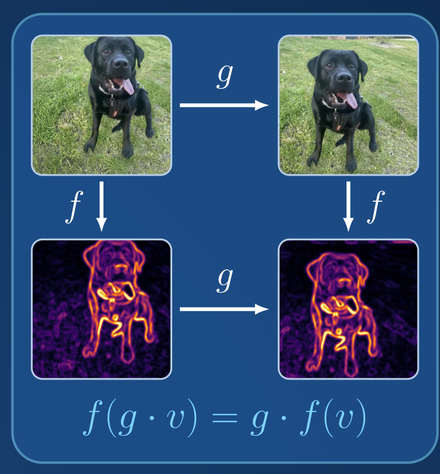
NeurReps 2026 · Mani Shayestehfar · The University of Sydney

Nonlinearities let different equivariant symmetry types talk to each other

Nonlinear Maps Between Irreducible Representations in Graph Neural Networks

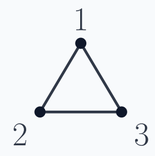


Why this matters

Linear equivariant layers keep symmetry types apart; only the activation lets them meet.

Graph networks respect relabelling of the vertices, so their features split into irreducible pieces: the **symmetry types**. By Schur's lemma, an equivariant *linear* layer never passes information between different types, so any interaction must come from the nonlinearity [1].

Example. S_3 permutes the coordinates of $\mathbb{R}^3$ (labelled vertices of a triangle).



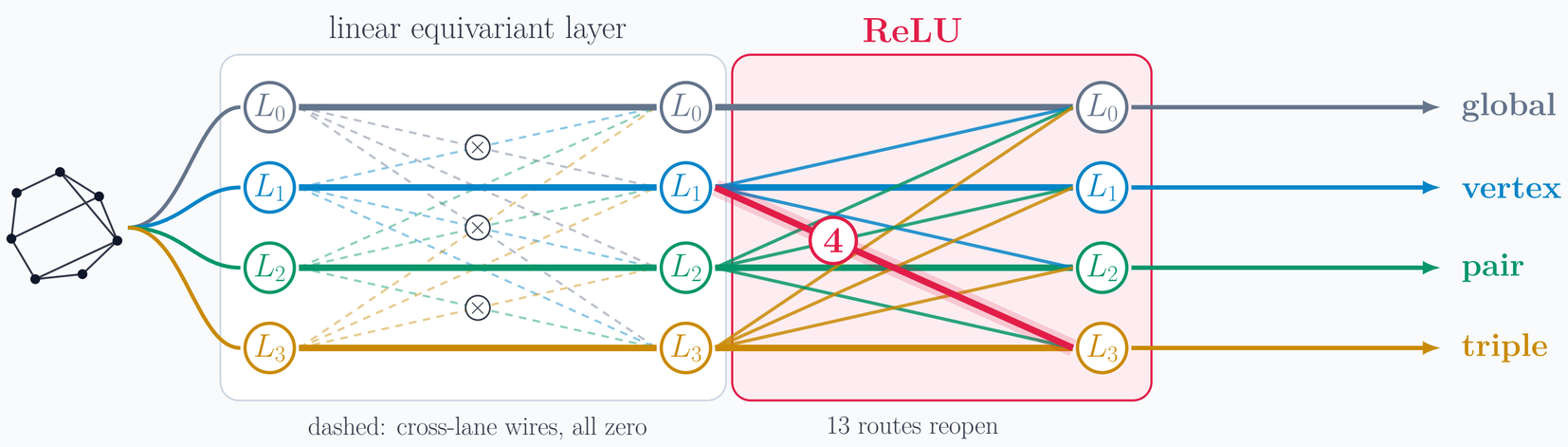
$$\mathbb{R}^3 = \underbrace{\mathbb{R}(1,1,1)}_{L_0\text{: constant}} \oplus \underbrace{\{x_1 + x_2 + x_3 = 0\}}_{L_1\text{: zero sum}}$$

$(-1,\ 0,\ 1)$ in L_1 (zero sum) —**ReLU**, $t \mapsto \max(t, 0)$→ $(0,\ 0,\ 1)$ sum is no longer 0 —average p_0→ $\frac{1}{3}(1,1,1)$ in L_0, and $\neq 0$

What to notice. linearly $\mathrm{Hom}_{S_3}(L_1, L_0) = 0$, yet after a ReLU a zero-sum vector has a nonzero average: a map $L_1 \to L_0$ exists only through the activation.

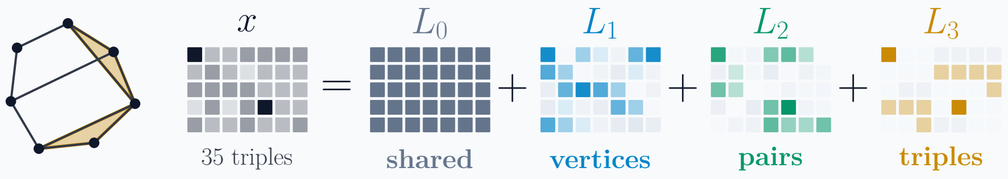
The idea

Split features into symmetry types, then map which wires the ReLU reopens, and at which degree.



What to notice. the linear layer passes each lane straight through; the ReLU reopens exactly 13 routes (vertex triples, 15–24 vertices). The global lane only feeds itself; vertex → triple opens only at degree 4.

a What the four lanes are



What to notice. a number on each vertex triple splits exactly into what all triples share, what vertices and pairs explain, and a genuinely triple-level rest: the two triangles.

b One route, three filters

	Symmetry allowed?	Coordinates produced?	Activation kept?	vertex → triple
$k = 1$	✗	–	–	closed
$k = 2$	✗	–	–	closed
$k = 3$	✓	✓	✗	removed by ReLU
$k = 4$	✓	✓	✓	**open**

What to notice. a route needs all three at the same degree. ReLU keeps no odd degree above 1, so vertex → triple is removed at $k = 3$ and opens at $k = 4$.

Theorem (each degree factors into the three filters) $\quad \Pi_k F_{j \leftarrow i} = a_k(\sigma; s_i)\, s_i^{-k}\, \Pi_k Q_k^{ji}, \quad Q_k^{ji} \in \mathrm{Hom}_G(\mathrm{Sym}^k L_i, L_j)$

The result

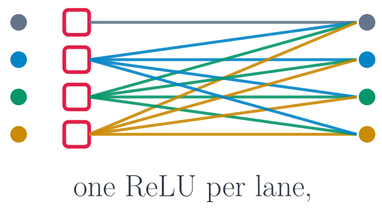
Keeping the routes separate gives **11% lower error** than an equal-size control, on a small benchmark.

Projector Control



all lanes mixed, then one shared ReLU

Specht Triple



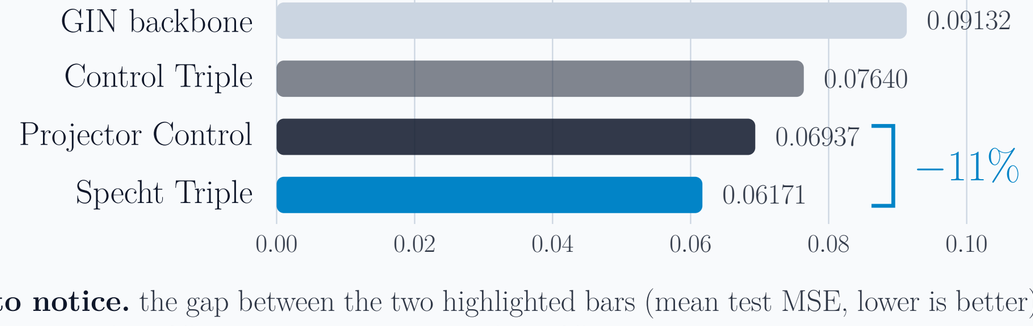
one ReLU per lane, 13 routes kept apart

Both: residuals on the same frozen GIN, same projectors, full linear equivariant layer, **884 parameters**. Only the nonlinearity differs.

Small benchmark. Synthetic graphs, targets: diameter and algebraic connectivity, frozen message-passing backbone.

−11% test error of Specht Triple vs **Projector Control**

−11.05% · 95% bootstrap CI [2.2%, 18.3%] · 7 of 10 seeds



What to notice. the gap between the two highlighted bars (mean test MSE, lower is better). Most of it is on diameter (−14%); algebraic connectivity (−2.6%) is unresolved.

Where this leads

Activation design: a bias restores routes ReLU removes.
Degree as an axis: own parameters per degree.
Beyond triples: larger subsets, other groups.
Triangulations: classifying manifolds.

[1] Gibson, Tubbenhauer, Williamson (2025) · [2] Shayestehfar (2026)

5-minute video watch on YouTube

Theory preprint arXiv:2609.25776

Code & data GitHub addendum



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Not here in person. Happy to chat online!